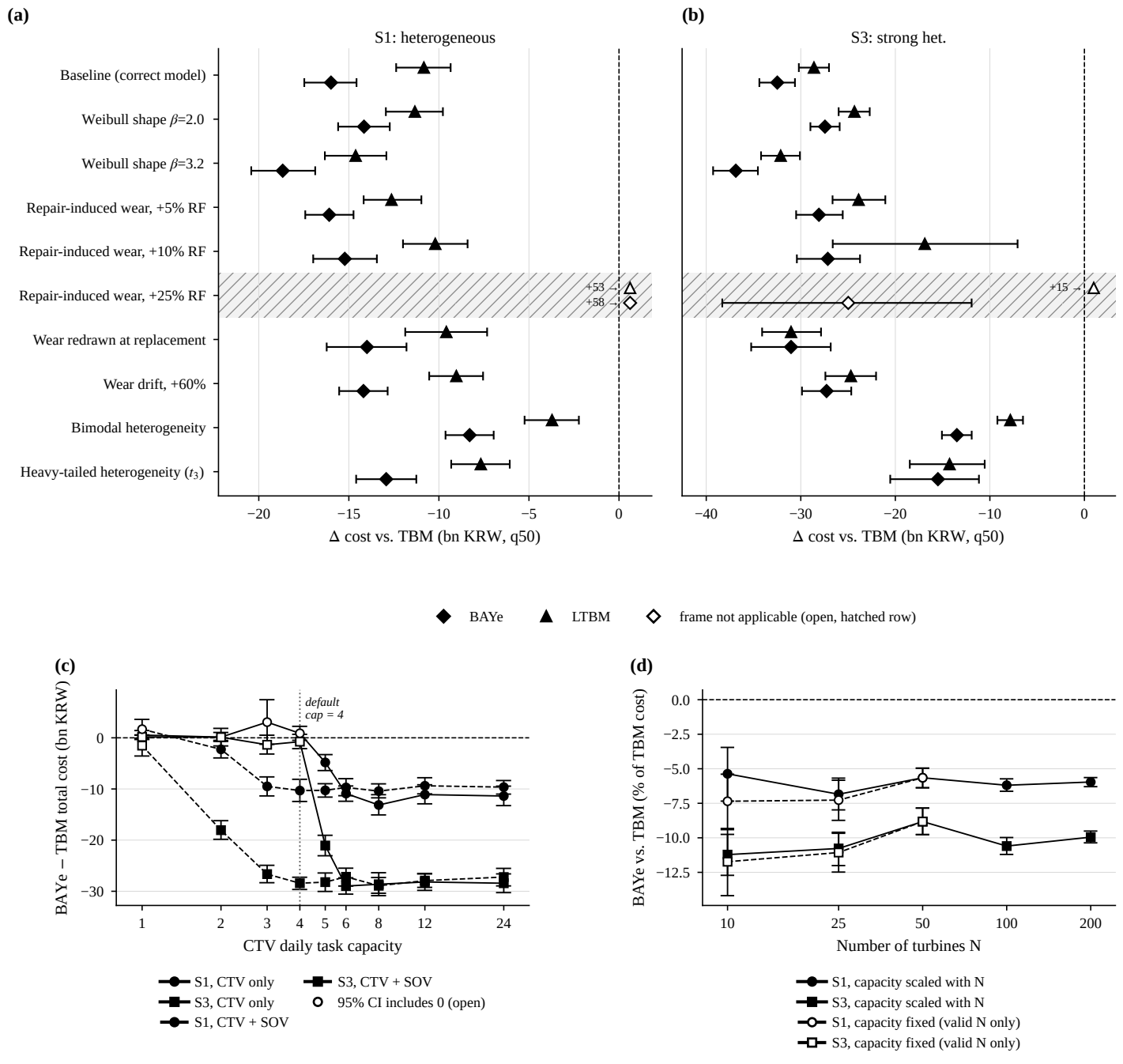

(a)
S1: heterogeneous
Baseline (correct model)
Weibull shape β=2.0
Weibull shape β=3.2
Repair-induced wear, +5% RF
Repair-induced wear, +10% RF
Repair-induced wear, +25% RF
Wear redrawn at replacement
Wear drift, +60%
Bimodal heterogeneity
Heavy-tailed heterogeneity (t3)
+53
+58
−20
−15
−10
−5
0
Δ cost vs. TBM (bn KRW, q50)
(b)
S3: strong het.
+15
−40
−30
−20
−10
0
Δ cost vs. TBM (bn KRW, q50)
BAYe
LTBM
frame not applicable (open, hatched row)
(c)
BAYe − TBM total cost (bn KRW)
default
cap = 4
0
−10
−20
−30
1
2
3
4
5
6
8
12
24
CTV daily task capacity
S1, CTV only
S3, CTV only
S1, CTV + SOV
S3, CTV + SOV
95% CI includes 0 (open)
(d)
BAYe vs. TBM (% of TBM cost)
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
10
25
50
100
200
Number of turbines N
S1, capacity scaled with N
S3, capacity scaled with N
S1, capacity fixed (valid N only)
S3, capacity fixed (valid N only)